PROGRAM:



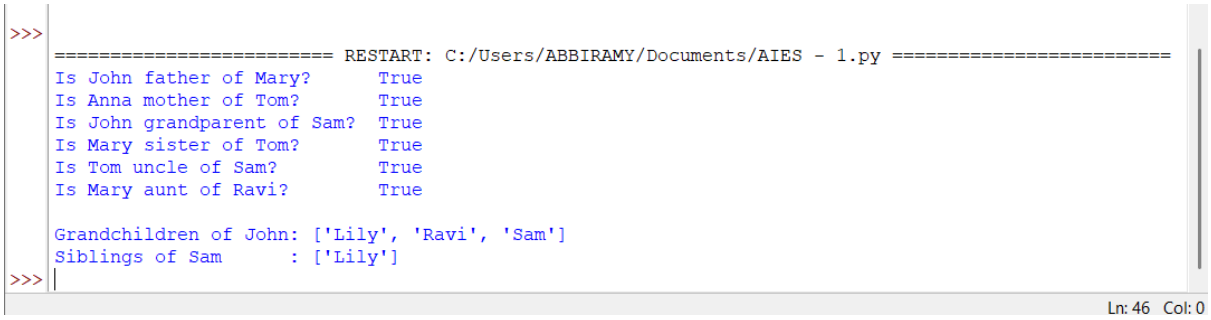
AIES - 1.py - C:/Users/ABBIRAMY/Documents/AIES - 1.py (3.14.2)

File Edit Format Run Options Window Help

```
parent = {
    ("John", "Mary"), ("John", "Tom"), ("Anna", "Mary"), ("Anna", "Tom"),
    ("Mary", "Sam"), ("Mary", "Lily"), ("Raj", "Sam"), ("Raj", "Lily"),
    ("Tom", "Ravi"),
}
male   = {"John", "Tom", "Raj", "Sam", "Ravi"}
female = {"Anna", "Mary", "Lily"}
people = {p for pair in parent for p in pair}
def father(x, y):      return (x, y) in parent and x in male
def mother(x, y):      return (x, y) in parent and x in female
def grandparent(x, y): return any((x, z) in parent and (z, y) in parent for z in people)
def sibling(x, y):     return x != y and any((p, x) in parent and (p, y) in parent for p in people)
def brother(x, y):     return sibling(x, y) and x in male
def sister(x, y):      return sibling(x, y) and x in female
def uncle(x, y):       return x in male and any(parent_of_y_sibling(x, y, p) for p in people)
def aunt(x, y):        return x in female and any(parent_of_y_sibling(x, y, p) for p in people)
def parent_of_y_sibling(x, y, p):
    return (p, y) in parent and sibling(x, p)
print("Is John father of Mary?     ", father("John", "Mary"))
print("Is Anna mother of Tom?      ", mother("Anna", "Tom"))
print("Is John grandparent of Sam? ", grandparent("John", "Sam"))
print("Is Mary sister of Tom?      ", sister("Mary", "Tom"))
print("Is Tom uncle of Sam?        ", uncle("Tom", "Sam"))
print("Is Mary aunt of Ravi?       ", aunt("Mary", "Ravi"))
print()
print("Grandchildren of John:", sorted(y for y in people if grandparent("John", y)))
print("Siblings of Sam      :", sorted(y for y in people if sibling("Sam", y)))
```

Ln: 19 Col: 0

OUTPUT:



```
>>>
========================= RESTART: C:/Users/ABBIRAMY/Documents/AIES - 1.py =========================
Is John father of Mary?      True
Is Anna mother of Tom?       True
Is John grandparent of Sam?  True
Is Mary sister of Tom?       True
Is Tom uncle of Sam?         True
Is Mary aunt of Ravi?        True

Grandchildren of John: ['Lily', 'Ravi', 'Sam']
Siblings of Sam      : ['Lily']
>>>
```

Ln: 46 Col: 0